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## Nickel-catalyzed cross-coupling of $\beta$ -carbonyl alkenyl pivalates with arylzinc chlorides†

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The nickel-catalyzed cross-coupling reaction of  $\beta$ -carbonyl alkenyl pivalates with arylzinc reagents generates 3-aryl-substituted  $\alpha,\beta$ -unsaturated carbonyl compounds via C–O bond cleavage. The reaction features mild reaction conditions, a wide scope of substrates, and good functional group tolerance.

### Introduction

$\alpha,\beta$ -Unsaturated carbonyl compounds are widely present in natural and bioactive molecules.<sup>1</sup> They are also used in the synthesis of pharmaceuticals, natural products, pesticides, and other useful materials.<sup>1,2</sup> A range of classical methods such as Wittig reaction,<sup>3</sup> Peterson olefination,<sup>4</sup> Horner–Wadsworth–Emmons reaction<sup>5</sup> and Claisen–Schmidt condensation<sup>6</sup> have been used to synthesize these compounds. In recent years, transition-metal-catalyzed reactions such as cross-couplings as versatile and highly effective synthetic methods have been developed to prepare these compounds or perform functionalization of  $\alpha,\beta$ -unsaturated carbonyl compounds.<sup>7–9</sup> For example, Bethi and Fernandes carried out one-pot synthesis of  $\alpha,\beta$ -unsaturated methyl ketones from homoallyl alcohols by sequential a  $\text{PdCl}_2/\text{CrO}_3$ -promoted Wacker process followed by an acid-mediated dehydration reaction.<sup>7a</sup> Cheng and co-workers developed a palladium-catalyzed dehydrogenative coupling reaction of saturated carbonyl compounds with aryl halides to prepare highly substituted  $\alpha,\beta$ -unsaturated carbonyl compounds.<sup>7b</sup> Li *et al.* reported Ru-catalyzed selective synthesis of (*E*)-2-arylchinnamaldehydes from (*E*)- $\beta$ -bromostyrenes and aryl aldehydes.<sup>7c</sup> Shi *et al.* reported  $\beta$ -alkylation of  $\alpha,\beta$ -unsaturated ketones or esters via Fe-catalyzed cross-coupling of alkenyl pivalates with alkyl Grignard reagents.<sup>8b</sup> In view of the importance of  $\alpha,\beta$ -unsaturated carbonyl compounds and inspired by the reported synthetic methods, we initiated a

study of nickel-catalyzed cross-coupling of  $\beta$ -carbonyl alkenyl pivalates with arylzinc reagents. Herein we report the results.

### Results and discussion

To screen catalysts, solvents, and other reaction conditions, reaction of 4-oxopent-2-en-2-yl pivalate (**1a**) with *p*- $\text{MeC}_6\text{H}_4\text{ZnCl}$  (**2**) was chosen as the model reaction. It has been reported that nickel can effectively catalyze C–O bond activation.<sup>10</sup> Hence we chose nickel complexes as pre-catalysts in this study.  $\text{Ni}(\text{PCy}_3)_2\text{Cl}_2$  was first tested and it led to 74% product yield when the reaction was run in THF at 30 °C for 24 h (Table 1, entry 1). Next, we examined the reaction in different solvents including  $\text{Et}_2\text{O}$ , dioxane, toluene, THF/NMP, THF/DMF and THF/DMA and found that toluene and 3 : 1 THF/NMP were also suitable solvents for this transformation besides THF (Table 1, entries 2–9). However, further tests showed that THF and toluene were not suitable for a wide scope of substrates. Hence screening of other conditions was performed in a 3 : 1 THF/NMP mixture. Other nickel complexes including  $\text{Ni}(\text{dppp})\text{Cl}_2$ ,  $\text{Ni}(\text{PPh}_3)_2\text{Cl}_2$ ,  $\text{Ni}(\text{DME})\text{Cl}_2$ , and a combination of  $\text{Ni}(\text{DME})\text{Cl}_2$  and TMEDA were tested and each of them was demonstrated to be less effective than  $\text{Ni}(\text{PCy}_3)_2\text{Cl}_2$  (Table 1, entries 10–13). Increasing or decreasing the loading amount of  $\text{Ni}(\text{PCy}_3)_2\text{Cl}_2$  resulted in a yield decrease (Table 1, entries 14 and 15). In the above reactions, the zinc reagent was prepared by reaction of *p*- $\text{MeC}_6\text{H}_4\text{MgBr}$  with an equimolar amount of  $\text{ZnCl}_2$  in the presence of 2 equiv. of LiCl. In the absence of a LiCl additive the coupling reaction gave 51% product yield. If the zinc reagent was prepared from *p*- $\text{MeC}_6\text{H}_4\text{Li}$  and  $\text{ZnCl}_2$ , the coupling reaction afforded only a trace amount of desired product (Table 1, entries 16 and 17). These experimental facts showed that both  $\text{Li}^+$  and  $\text{Mg}^{2+}$  ions played important roles in the catalytic transformation. It has been reported that action of  $\text{ArZnX}$  with  $\text{MgX}_2$  led to formation of ionic zincates which have higher reactivity than  $\text{ArZnX}$ .<sup>11</sup> The role of LiX may involve increasing the solubility of the zinc

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**Table 1** Screening of catalyst and optimization of reaction conditions<sup>a</sup>

| Entry           | Catalyst                                           | Solvent           | T (°C) | Yield <sup>b</sup> (%) |
|-----------------|----------------------------------------------------|-------------------|--------|------------------------|
| 1               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF               | 30     | 74                     |
| 2               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | Et <sub>2</sub> O | 30     | 53                     |
| 3               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | Dioxane           | 30     | 40                     |
| 4               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | Toluene           | 30     | 78                     |
| 5               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (1 : 1)   | 30     | 69                     |
| 6               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (2 : 1)   | 30     | 72                     |
| 7               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | 76                     |
| 8               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/DMF (3 : 1)   | 30     | 54                     |
| 9               | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/DMA (3 : 1)   | 30     | 45                     |
| 10              | Ni(dppp)Cl <sub>2</sub>                            | THF/NMP (3 : 1)   | 30     | 69                     |
| 11              | Ni(PPh <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | 66                     |
| 12              | Ni(DME)Cl <sub>2</sub>                             | THF/NMP (3 : 1)   | 30     | 69                     |
| 13 <sup>c</sup> | Ni(DME)Cl <sub>2</sub>                             | THF/NMP (3 : 1)   | 30     | 69                     |
| 14 <sup>d</sup> | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | 69                     |
| 15 <sup>e</sup> | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | 64                     |
| 16 <sup>f</sup> | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | 51                     |
| 17 <sup>g</sup> | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 30     | Trace                  |
| 18              | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 40     | 67                     |
| 19              | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 25     | 74                     |
| 20              | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 20     | 73                     |
| 21              | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 10     | 49                     |
| 22 <sup>h</sup> | Ni(PCy <sub>3</sub> ) <sub>2</sub> Cl <sub>2</sub> | THF/NMP (3 : 1)   | 25     | 60                     |

<sup>a</sup>The reactions were carried out on a 0.3 mmol scale, and 1.5 equiv. of *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl and 2 cm<sup>3</sup> of solvent were employed. Unless otherwise specified, *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl was prepared from *p*-MeC<sub>6</sub>H<sub>4</sub>MgBr and 1 equiv. of ZnCl<sub>2</sub> in the presence of 2 equiv. of LiCl. <sup>b</sup>Isolated yield.

<sup>c</sup>7.5 mol% of TMEDA was employed as the ligand. <sup>d</sup>2.5 mol% Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> was employed. <sup>e</sup>7.5 mol% Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> was employed. <sup>f</sup>*p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl was prepared from *p*-MeC<sub>6</sub>H<sub>4</sub>MgBr and 1 equiv. of ZnCl<sub>2</sub>.

<sup>g</sup>*p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl was prepared from *p*-MeC<sub>6</sub>H<sub>4</sub>Li and 1 equiv. of ZnCl<sub>2</sub>.

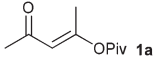
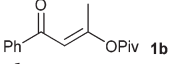
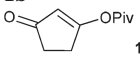
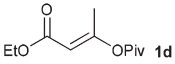
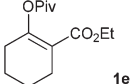
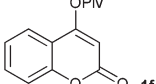
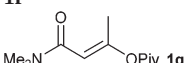
<sup>h</sup>The reaction time was 12 h.

reagents through forming a trimetallic adduct and producing reactive zincates.<sup>12</sup> The temperature effect was next tested. The reaction in THF/NMP at 40 °C gave a slightly lower yield and at 25 °C and 20 °C gave comparable yields to that at 30 °C. When the reaction was run at 10 °C, the product yield markedly reduced (Table 1, entries 18–21). We also noticed that shortening the reaction time led to a yield decrease (Table 1, entry 22). Finally, the reaction mixture obtained under the optimized conditions was examined using GC-MS to identify the by-products. The homocoupling species of the zinc reagent, 4,4'-dimethylbiphenyl, was a main by-product. Small amounts of 4,5-dimethylocta-3,5-diene-2,7-dione and 4,4-di-*p*-tolylpentan-2-one were also observed. The former was the homocoupling species of **1a** and the latter probably resulted from conjugate addition of **3a** with the zinc reagent.

Next, the substrate scope of the cross-coupling reaction was examined. The reactions in THF often led to slightly lower yields than those in THF/NMP (3 : 1) mixture in a series of tests. Toluene was only suitable as the reaction solvent for some substrates. Hence next tests were mostly carried out in a 3 : 1 mixture of THF and NMP. We have demonstrated that 4-oxopent-2-en-2-yl pivalate (**1a**) reacted with *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl to afford a cross-coupling product in good yield (Table 1). Further

tests showed that other electron-rich zinc reagents including *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl and *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl also reacted smoothly with **1a** to give corresponding cross-coupling products in good yields (Table 2, entries 1 and 2). Electron-poor arylzinc chloride, *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl, reacted with **1a** under standard conditions to form desired product in 52% yield. Elevated reaction temperature resulted in a slight yield increase (59%) (Table 2, entry 3). The relatively low yield was ascribed to the weak nucleophilic power of *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl. Reaction of *o*-MeC<sub>6</sub>H<sub>4</sub>ZnCl with **1a** afforded 4-*o*-tolylpent-3-en-2-one in 89% yield, including 55% of (*E*)-isomer and 34% of (*Z*)-isomer which can be separated by column chromatography (Table 2, entry 4). It seems that the steric hindrance of the *o*-tolyl group did not affect the reactivity of the zinc reagent and *o*-MeC<sub>6</sub>H<sub>4</sub>ZnCl even led to higher product yield than *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl. This is probably because steric hindrance of the *o*-tolyl group prevented the homo-coupling side reaction of the zinc reagent. 2-Thienylzinc chloride as a representative of a heteroarylzinc reagent was demonstrated to react with **1a** under standard conditions, giving 4-(thiophen-2-yl)pent-3-en-2-one in 58% yield (Table 2, entry 5). 4-Oxo-4-phenylbut-2-en-2-yl pivalate (**1b**) exhibited higher reactivity than **1a** when it reacted with either electron-rich or electron-poor arylzinc reagents, giving corresponding cross-coupling products in 71–90% yields (Table 2, entries 6–9). The high reactivity of **1b** probably results from the existence of an extended  $\pi$ -conjugated system. 3-Oxocyclopent-1-enyl pivalate (**1c**) displayed a comparable reactivity to **1a**. Its reactions with *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl, *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl, and *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl resulted in desired products in 75–88% yields (Table 2, entries 10–12). *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl still showed relatively low reactivity in the reaction with **1c**, leading to the cross-coupling product in 50% yield (Table 2, entry 13). 2-Furylzinc chloride was demonstrated to react with **1c** to afford 3-(furan-2-yl)cyclopent-2-enone in 45% yield (Table 2, entry 14). Reaction of 2-thienylzinc chloride with **1c** formed desired product in 89% yield (Table 2, entry 15), which was much higher than that obtained from its reaction with **1a**. We noticed that the reactions of *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl, *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl, and 2-thienylzinc chloride with **1c** proceeded better in toluene than in a 3 : 1 mixture of THF and NMP, and the reactions of *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl, 2-furylzinc chloride, and 2-thienylzinc chloride required a higher reaction temperature (50 °C) to achieve optimal results. Ethyl 3-(pivaloyloxy)but-2-enoate (**1d**) exhibited higher reactivity than **1a**. In most reactions tested its reactivity was comparable to that of **1b**. For example, reaction of **1d** with *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl, *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl, *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl, and *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl gave yields of desired products comparable to those of the corresponding reactions of **1b** (Table 2, entries 16–19). Reaction of **1d** with *o*-MeC<sub>6</sub>H<sub>4</sub>ZnCl generated ethyl 3-*o*-tolylbut-2-enoate in 92% yield (Table 2, entry 20), which is consistent with the result for the reaction of **1a** with *o*-MeC<sub>6</sub>H<sub>4</sub>ZnCl. Compound **1d** also reacted smoothly with 2-furylzinc chloride and 2-thienylzinc chloride at 50 °C to afford the desired products in 69% and 82% yields, respectively (Table 2, entries 21 and 22). Ethyl 2-(pivaloyloxy)cyclohex-1-ene-1-carboxylate (**1e**) showed a lower

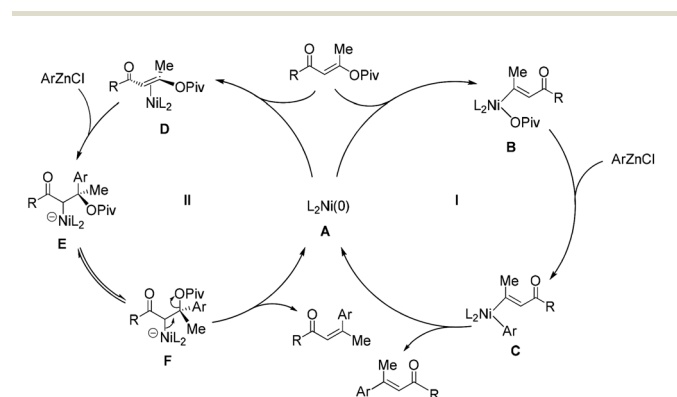
**Table 2** Cross-coupling of  $\beta$ -carbonyl alkenyl pivalates with arylzinc chlorides<sup>a</sup>

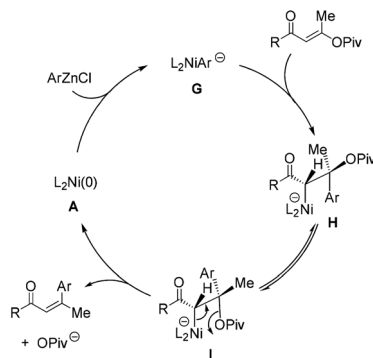
| $  \begin{array}{c}  \text{R}^1-\text{C}(=\text{O})-\text{CH}=\text{CH}-\text{OPiv} + \text{ArZnCl} \xrightarrow[\text{THF/NMP (3:1), 25}^\circ\text{C, 24 h}]{\text{Ni(PCy}_3)_2\text{Cl}_2 \text{ (5 mol\%)}} \\  \text{1a-1g} \hspace{10em} \text{R}^1-\text{C}(=\text{O})-\text{CH}=\text{CH}-\text{Ar}  \end{array}  $ |                                                                                     |                                                          |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------|
| Entry                                                                                                                                                                                                                                                                                                                       | Alkenyl-OPiv                                                                        | Ar                                                       | Yield <sup>b</sup> (%) |
| 1                                                                                                                                                                                                                                                                                                                           |    | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 84                     |
| 2                                                                                                                                                                                                                                                                                                                           | <b>1a</b>                                                                           | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 77                     |
| 3                                                                                                                                                                                                                                                                                                                           | <b>1a</b>                                                                           | <i>p</i> -CF <sub>3</sub> C <sub>6</sub> H <sub>4</sub>  | 52 (59 <sup>c</sup> )  |
| 4                                                                                                                                                                                                                                                                                                                           | <b>1a</b>                                                                           | <i>o</i> -MeC <sub>6</sub> H <sub>4</sub>                | 89 <sup>d</sup>        |
| 5                                                                                                                                                                                                                                                                                                                           | <b>1a</b>                                                                           | 2-Thienyl                                                | 58                     |
| 6                                                                                                                                                                                                                                                                                                                           | <b>1a</b>                                                                           | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 86                     |
| 7                                                                                                                                                                                                                                                                                                                           |    | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 90                     |
| 8                                                                                                                                                                                                                                                                                                                           | <b>1b</b>                                                                           | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 89                     |
| 9                                                                                                                                                                                                                                                                                                                           | <b>1b</b>                                                                           | <i>p</i> -CF <sub>3</sub> C <sub>6</sub> H <sub>4</sub>  | 71 <sup>c</sup>        |
| 10                                                                                                                                                                                                                                                                                                                          |    | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 88                     |
| 11                                                                                                                                                                                                                                                                                                                          | <b>1c</b>                                                                           | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 82                     |
| 12                                                                                                                                                                                                                                                                                                                          | <b>1c</b>                                                                           | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 75 <sup>e</sup>        |
| 13                                                                                                                                                                                                                                                                                                                          | <b>1c</b>                                                                           | <i>p</i> -CF <sub>3</sub> C <sub>6</sub> H <sub>4</sub>  | 50 <sup>c,e</sup>      |
| 14                                                                                                                                                                                                                                                                                                                          | <b>1c</b>                                                                           | 2-Furyl                                                  | 45 <sup>c</sup>        |
| 15                                                                                                                                                                                                                                                                                                                          | <b>1c</b>                                                                           | 2-Thienyl                                                | 89 <sup>c,e</sup>      |
| 16                                                                                                                                                                                                                                                                                                                          |    | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 93                     |
| 17                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 83                     |
| 18                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 85                     |
| 19                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | <i>p</i> -CF <sub>3</sub> C <sub>6</sub> H <sub>4</sub>  | 74 <sup>c</sup>        |
| 20                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | <i>o</i> -MeC <sub>6</sub> H <sub>4</sub>                | 92                     |
| 21                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | 2-Furyl                                                  | 69 <sup>c</sup>        |
| 22                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | 2-Thienyl                                                | 82 <sup>c</sup>        |
| 23                                                                                                                                                                                                                                                                                                                          | <b>1d</b>                                                                           | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 77                     |
| 24                                                                                                                                                                                                                                                                                                                          |  | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 74                     |
| 25                                                                                                                                                                                                                                                                                                                          | <b>1e</b>                                                                           | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 54 (61 <sup>c</sup> )  |
| 26                                                                                                                                                                                                                                                                                                                          | <b>1e</b>                                                                           | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 88                     |
| 27                                                                                                                                                                                                                                                                                                                          |  | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 78                     |
| 28                                                                                                                                                                                                                                                                                                                          | <b>1f</b>                                                                           | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 70                     |
| 29                                                                                                                                                                                                                                                                                                                          | <b>1f</b>                                                                           | <i>p</i> -CF <sub>3</sub> C <sub>6</sub> H <sub>4</sub>  | 73 <sup>c</sup>        |
| 30                                                                                                                                                                                                                                                                                                                          | <b>1f</b>                                                                           | 2-Furyl                                                  | 55                     |
| 31                                                                                                                                                                                                                                                                                                                          | <b>1f</b>                                                                           | 2-Thienyl                                                | 69 <sup>c</sup>        |
| 32                                                                                                                                                                                                                                                                                                                          | <b>1f</b>                                                                           | <i>p</i> -Me <sub>2</sub> NC <sub>6</sub> H <sub>4</sub> | 99 <sup>f</sup>        |
| 33                                                                                                                                                                                                                                                                                                                          |  | <i>p</i> -MeOC <sub>6</sub> H <sub>4</sub>               | 99 <sup>f</sup>        |
| 34                                                                                                                                                                                                                                                                                                                          | <b>1g</b>                                                                           | <i>p</i> -MeC <sub>6</sub> H <sub>4</sub>                | 90 <sup>f</sup>        |

<sup>a</sup> Unless otherwise specified, the reactions were carried out on a 0.3 mmol scale according to the conditions indicated by above equation; 1.5 equiv. of arylzinc chlorides and 2 cm<sup>3</sup> solvent were employed. *p*-RC<sub>6</sub>H<sub>4</sub>ZnCl was prepared from *p*-RC<sub>6</sub>H<sub>4</sub>MgBr and 1 equiv. of ZnCl<sub>2</sub> in the presence of 2 equiv. of LiCl; 2-furylzinc chloride and 2-thienylzinc chloride were prepared by reaction of 2-furyllithium or 2-thienyllithium with 1 equiv. of ZnCl<sub>2</sub> in the presence of 1 equiv. of LiCl and 1 equiv. of MgBr<sub>2</sub>. <sup>b</sup> Isolated yield. <sup>c</sup> The reaction was run at 50 °C. <sup>d</sup> The product includes 55% (*E*)-4-(*o*-tolyl)pent-3-en-2-one and 34% (*Z*)-4-(*o*-tolyl)pent-3-en-2-one. <sup>e</sup> Toluene was used as the reaction solvent. <sup>f</sup> A mixture of *Z* and *E* isomers was obtained.

reactivity than **1d**. It exhibited a similar reactivity to **1a** when *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl and *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl were employed as the nucleophilic reagents (Table 2, entries 23 and 24). However, reaction of **1e** with *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl under standard conditions resulted in relatively low yield (54%) for unclear reasons. Raising the reaction temperature (50 °C) led to only a small increase of the product yield (Table 2, entry 25). 2-Oxo-2*H*-chromen-4-yl pivalate (**1f**) as a lactone derivative displayed a similar reactivity to **1a** when it reacted with electron-rich zinc reagents including *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl, *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl and *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl and showed a slightly higher reactivity than **1a** when it reacted with the electron-poor zinc reagent *p*-CF<sub>3</sub>C<sub>6</sub>H<sub>4</sub>ZnCl, and heteroarylzinc reagent 2-thienylzinc chloride (Table 2, entries 26–31). The lower reactivity of both **1e** and **1f** than **1d** was probably because the former have larger steric hindrance. 4-(Dimethylamino)-4-oxobut-2-en-2-yl pivalate (**1g**) exhibited excellent reactivity. Its reactions with *p*-Me<sub>2</sub>NC<sub>6</sub>H<sub>4</sub>ZnCl and *p*-MeOC<sub>6</sub>H<sub>4</sub>ZnCl gave desired products in quantitative yields, and with *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl afforded the cross-coupling product in 90% yield (Table 2, entries 32–34). In addition, each reaction of **1g** tested gave a mixture of (*Z*)- and (*E*)-isomers of the cross-coupling products.

The Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub>-catalyzed reaction of 4-oxopent-2-en-2-yl pivalate with *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl was not affected by 1,1-diphenylethene additive. The reaction run under the optimized conditions in the presence of 1,1-diphenylethene (1 equiv.) gave the desired product in 74% yield. This ruled out the possibility of a radical process. Reaction of 4-oxopent-2-en-2-yl pivalate with *p*-MeC<sub>6</sub>H<sub>4</sub>ZnCl could not occur in the absence of Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub>, which means that a direct nucleophilic substitution was impossible. When a combination of Ni(COD)<sub>2</sub> (5 mol%) and PCy<sub>3</sub> (10 mol%) was employed to replace Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> as the catalyst, the reaction afforded cross-coupling product in 75% yield. This experimental fact showed that the catalytically active species might be a Ni(0) complex. Based on the above experimental results and the reported mechanism of nickel-catalyzed cross-coupling reactions,<sup>9a,c</sup> a possible catalytic cycle is outlined in Scheme 1 (catalytic cycle I). Thus, reaction of Ni(PCy<sub>3</sub>)<sub>2</sub>Cl<sub>2</sub> with arylzinc chloride generates L<sub>2</sub>Ni(0) (**A**) which is an active catalyst. Oxidative addition of  $\beta$ -carbonyl alkenyl pivalate to L<sub>2</sub>Ni(0) forms L<sub>2</sub>Ni(OPiv)(C(Me)=CHC(O)R) (**B**).

**Scheme 1** Proposed catalytic cycle.



**Scheme 2** Another possible pathway to generate (*Z*)-isomer.

Substitution of OPiv from the metal center in complex **B** by the arylzinc reagent results in  $L_2Ni(Ar)(C(Me)=CHC(O)R)$  (**C**). The intermediate **C** undergoes reductive elimination to give cross-coupling product and regenerating  $L_2Ni(0)$ . In the process an (*E*)-isomeric product should be formed from (*E*)-β-carbonyl alkenyl pivalate. However, in the reactions of **1a** with *o*-MeC<sub>6</sub>H<sub>4</sub>ZnCl as well as **1g** with various zinc reagents, a mixture of (*Z*)- and (*E*)-isomers was obtained in each case. The <sup>1</sup>H NMR spectra of starting material **1a**, and the products (*Z*)- and (*E*)-4-(*o*-tolyl)pent-3-en-2-one showed that they were stable in CDCl<sub>3</sub> solution in 37 hours, no isomerized species being observed in each case (see Fig. S1–S3 in the ESI†). GC monitoring of the reaction process also showed that (1) no isomerized species of compound **1a** was detected; and (2) both (*Z*)- and (*E*)-4-(*o*-tolyl)pent-3-en-2-one appeared in the early stage of the reaction, and the ratio of the (*Z*)- and (*E*)-isomers was approximately constant as the reaction progressed (see Fig. S7–S10 in the ESI†). These results imply that there might be a competitive reaction process. This process is also nickel-catalyzed because the cross-coupling reaction cannot occur in the absence of a nickel catalyst. The competitive catalytic cycle is proposed in Scheme 1 (cycle II). In this cycle, interaction of  $L_2Ni(0)$  with (*E*)-β-carbonyl alkenyl pivalate forms an alkene-nickel coordination complex **D**. Addition of ArZnCl to the C–C double bond of **D** from the opposite side of the  $L_2Ni$  unit generates the intermediate **E**.<sup>13,14</sup> **E** can convert to conformation **F** in which the OPiv group and the  $NiL_2$  moiety are anti-periplanar. Anti-elimination of OPiv<sup>−</sup> and  $NiL_2$  from **F** results in (*Z*)-RC(O)CH=C(Me)Ar.<sup>14</sup> However, we cannot rule out another pathway to generate the (*Z*)-isomer as shown in Scheme 2. Thus, interaction of ArZnCl with  $L_2Ni(0)$  forms an ate complex  $L_2NiAr^−$  (**G**). *syn* insertion of the C–C double bond of (*E*)-β-carbonyl alkenyl pivalate into the Ni–C<sub>Ar</sub> bond of the ate complex forms the intermediate **H**. Bond rotation of **H** results in **I**. *syn*-Elimination of OPiv<sup>−</sup> and  $NiL_2$  from **I** produces (*Z*)-RC(O)CH=C(Me)Ar.

## Conclusions

We have performed a nickel-catalyzed cross-coupling of β-carbonyl alkenyl pivalates with arylzinc chlorides under mild

reaction conditions. β-Arylated α,β-unsaturated ketones, esters, and amides were synthesized *via* the reaction. Both acyclic and cyclic β-carbonyl alkenyl pivalates were suitable for this transformation. Electron-rich and electron-poor arylzinc chlorides, 2-furylzinc chloride and 2-thienylzinc chloride can be employed as the nucleophilic reagents. Due to easy preparation of β-carbonyl alkenyl carboxylates from 1,3-dicarbonyl compounds, this approach provides a convenient route to synthesize β-aryl-substituted α,β-unsaturated carbonyl compounds from widely available 1,3-dicarbonyl compounds. Studies are in progress to expand the scope of the reaction to alkyl- and alkenylzinc reagents.

## Experimental

All air- or moisture-sensitive manipulations were performed under nitrogen using standard Schlenk techniques. Toluene, THF and Et<sub>2</sub>O were purified by J.C. Meyer Phoenix Solvent Systems. 1,4-Dioxane was distilled over sodium under nitrogen. NMP, DMF and DMA were dried over 4 Å molecular sieves, fractionally distilled under reduced pressure, and stored under a nitrogen atmosphere. Ethyl 2-(pivaloyloxy)cyclohex-1-ene-1-carboxylate (**1e**) and 2-oxo-2H-chromen-4-yl pivalate (**1f**) were prepared according to literature procedures.<sup>8b</sup> Other β-carbonyl alkenyl pivalates were prepared by similar methods. Grignard reagents<sup>15</sup> and lithium reagents<sup>16</sup> were prepared according to reported methods. Arylzinc chlorides were prepared from ZnCl<sub>2</sub> and 1 equiv. of ArMgBr in the presence of 2 equiv. of LiCl. 2-Furylzinc chloride and 2-thiophenylzinc chloride were prepared by reaction of 2-furyllithium or 2-thiophenyllithium with 1 equiv. of ZnCl<sub>2</sub> in the presence of 1 equiv. of LiCl and 1 equiv. of MgBr<sub>2</sub>. Other solvents and chemicals were obtained from commercial vendors and used as received. NMR spectra were recorded with a Bruker av400 NMR spectrometer at ambient temperature. The chemical shifts of <sup>1</sup>H NMR spectra were referenced to TMS, and the chemical shifts of <sup>13</sup>C NMR spectra were referenced to internal solvent resonances. High-resolution mass spectra (HR-MS) were acquired with a Thermo Fisher LTQ Orbitrap XL mass spectrometer in ESI mode using an Orbitrap mass analyzer.

### Synthesis of β-carbonyl alkenyl pivalates

**(*E*)-4-Oxopent-2-en-2-yl pivalate (1a).** Pivaloyl chloride (45 mmol) was added dropwise to a stirred solution of 2,4-pentanedione (30 mmol) and NEt<sub>3</sub> (60 mmol) in dry CH<sub>2</sub>Cl<sub>2</sub> (30 cm<sup>3</sup>) at 0 °C. The solution was stirred at room temperature for 6 h. The resulting mixture was successively washed with water (20 cm<sup>3</sup>) and saturated NH<sub>4</sub>Cl solution (20 cm<sup>3</sup>), dried over anhydrous Na<sub>2</sub>SO<sub>4</sub>, and evaporated to dryness by rotary evaporation. The residue was purified by column chromatography (eluted with 100 : 1 petroleum ether/EtOAc) to yield (*E*)-4-oxopent-2-en-2-yl pivalate as a light yellow oil, yield 2.27 g (41%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 1.27 (s, 9H), 2.22 (s, 3H), 2.30 (d, *J* = 0.7 Hz, 3H), 6.04 (s, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 18.60, 27.04, 32.26, 39.26, 116.32, 163.30, 176.10,



197.53. HR-MS:  $m/z$  207.09880  $[M + Na]^+$ ; calcd for  $C_{10}H_{16}O_3Na$ , 207.09917.

The same procedure as for (*E*)-4-oxopent-2-en-2-yl pivalate was used to prepare the other  $\beta$ -carbonyl alkenyl pivalates.

**(*E*)-4-Oxo-4-phenylbut-2-en-2-yl pivalate (1b).** Eluent: petroleum ether/EtOAc (40 : 1 v/v). Yellow oil, yield 0.67 g (54%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  1.31 (s, 9H), 2.39 (s, 3H), 6.75 (s, 1H), 7.46 (t,  $J$  = 7.6 Hz, 2H), 7.55 (t,  $J$  = 7.3 Hz, 1H), 7.90–7.95 (m, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  19.03, 27.10, 39.35, 113.47, 128.27, 128.69, 132.92, 138.83, 164.50, 176.14, 190.51. HR-MS:  $m/z$  269.11412  $[M + Na]^+$ ; calcd for  $C_{15}H_{18}O_3Na$ , 269.11482.

**3-Oxocyclopent-1-en-1-yl pivalate (1c).** Eluent: petroleum ether/EtOAc (20 : 1 v/v). Light yellow solid, yield 1.50 g (82%); m.p. 41–43 °C.  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  1.31 (s, 9H), 2.43–2.51 (m, 2H), 2.72–2.82 (m, 2H), 6.19–6.24 (m, 1H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  26.88, 28.76, 33.37, 39.72, 116.44, 174.10, 180.17, 206.98. HR-MS:  $m/z$  183.10204  $[M + H]^+$ ; calcd for  $C_{10}H_{15}O_3$ , 183.10157.

**Ethyl (*E*)-3-(pivaloyloxy)but-2-enoate (1d).**<sup>17</sup> Eluent: petroleum ether/EtOAc (200 : 1 v/v). Colorless oil, yield 1.94 g (45%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  1.26 (s, 9H), 1.28 (t,  $J$  = 7.2 Hz, 3H), 2.33 (d,  $J$  = 0.8 Hz, 3H), 4.18 (q,  $J$  = 7.1 Hz, 2H), 5.63 (q,  $J$  = 0.8 Hz, 1H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  13.32, 17.05, 25.99, 38.15, 59.21, 108.94, 163.48, 165.08, 174.95.

**(*E*)-4-(Dimethylamino)-4-oxobut-2-en-2-yl pivalate (1g).** Eluent: petroleum ether/EtOAc (3 : 1 v/v). Light yellow oil, yield 0.586 g (55%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  1.25 (s, 9H), 1.99 (d,  $J$  = 0.8 Hz, 3H), 2.92 (s, 3H), 3.01 (s, 3H), 5.68 (s, 1H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  20.12, 26.81, 34.48, 37.67, 38.82, 110.38, 152.04, 165.05, 175.62. HR-MS:  $m/z$  214.14430  $[M + H]^+$ ; calcd for  $C_{11}H_{20}NO_3$ , 214.14377.

### General procedure for the cross-coupling of $\beta$ -carbonyl alkenyl pivalate with arylzinc chlorides

A Schlenk tube was charged with  $\beta$ -carbonyl alkenyl pivalate (0.3 mmol), NMP (0.5 cm<sup>3</sup>), and  $Ni(PCy_3)_2Cl_2$  (0.015 mmol). To the stirred mixture was added dropwise  $ArZnCl$  solution (1.5 cm<sup>3</sup>, 0.3 M solution in THF, 0.45 mmol) by syringe. The reaction mixture was stirred at 25 °C for 24 h. Then 30%  $NH_4Cl$  solution (15 cm<sup>3</sup>) was added. The mixture was extracted with ethyl acetate (3  $\times$  10 cm<sup>3</sup>). The combined organic phases were dried over anhydrous  $Na_2SO_4$ , concentrated by rotary evaporation, and purified by column chromatography on silica gel.

**(*E*)-4-(*p*-Tolyl)pent-3-en-2-one.**<sup>18</sup> Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow oil, yield 38.6 mg (74%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.29 (s, 3H), 2.37 (s, 3H), 2.53 (d,  $J$  = 1.0 Hz, 3H), 6.51 (s, 1H), 7.19 (d,  $J$  = 8.0 Hz, 2H), 7.40 (d,  $J$  = 8.2 Hz, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  18.30, 21.31, 32.36, 123.80, 126.49, 129.33, 139.42, 139.54, 153.94, 199.06.

**(*E*)-4-(4-(Dimethylamino)phenyl)pent-3-en-2-one.** Eluent: petroleum ether/EtOAc (60 : 1 v/v). Yellow solid, yield 51.5 mg (84%); m.p. 112–114 °C.  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.27 (s, 3H), 2.54 (d,  $J$  = 0.8 Hz, 3H), 3.00 (s, 6H), 6.52 (s, 1H), 6.68 (d,  $J$  = 9.0 Hz, 2H), 7.47 (d,  $J$  = 9.0 Hz, 2H).  $^{13}C$  NMR (101 MHz,

$CDCl_3$ ):  $\delta$  18.20, 32.43, 55.51, 114.02, 122.98, 128.03, 134.63, 153.51, 160.69, 199.00. HR-MS:  $m/z$  204.13882  $[M + H]^+$ ; calcd for  $C_{13}H_{18}NO$ , 204.13829.

**(*E*)-4-(4-Methoxyphenyl)pent-3-en-2-one.**<sup>19,20</sup> Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow solid, yield 44 mg (77%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.29 (s, 3H), 2.53 (d,  $J$  = 1.1 Hz, 3H), 3.84 (s, 3H), 6.50 (s, 1H), 6.91 (d,  $J$  = 8.9 Hz, 2H), 7.48 (d,  $J$  = 8.9 Hz, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  18.19, 32.42, 55.49, 114.01, 122.96, 128.02, 134.62, 153.49, 160.68, 198.98.

**(*E*)-4-(4-(Trifluoromethyl)phenyl)pent-3-en-2-one.**<sup>20,21</sup> Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow oil, yield 40.5 mg (59%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.32 (s, 3H), 2.54 (s, 3H), 6.52 (s, 1H), 7.58 (d,  $J$  = 8.4 Hz, 2H), 7.64 (d,  $J$  = 8.3 Hz, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  18.43, 32.35, 124.07 (q,  $J$  = 273.1 Hz), 125.62 (q,  $J$  = 3.8 Hz), 126.02, 126.95, 130.94 (q,  $J$  = 32.7 Hz), 146.22, 152.16, 198.84.

**(*E*)-4-(*o*-Tolyl)pent-3-en-2-one.** Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow oil, yield 29 mg (55%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.26 (s, 3H), 2.29 (s, 3H), 2.41 (d,  $J$  = 1.4 Hz, 3H), 6.16 (q,  $J$  = 1.3 Hz, 1H), 7.07 (d,  $J$  = 7.1 Hz, 1H), 7.15–7.24 (m, 3H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  19.89, 21.35, 32.20, 125.86, 126.73, 127.19, 127.83, 130.56, 134.02, 144.19, 156.76, 199.10.

**(*Z*)-4-(*o*-Tolyl)pent-3-en-2-one.** Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow oil, yield 18 mg (34%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  1.70 (s, 3H), 2.13 (d,  $J$  = 1.4 Hz, 3H), 2.24 (s, 3H), 6.20 (q,  $J$  = 1.4 Hz, 1H), 6.98–7.03 (m, 1H), 7.17–7.25 (m, 3H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  19.22, 27.58, 29.66, 126.22, 126.61, 127.92, 129.50, 130.45, 133.62, 141.04, 154.26, 199.31. HR-MS:  $m/z$  175.11154  $[M + H]^+$ ; calcd for  $C_{12}H_{16}O$ , 175.11174.

**(*E*)-4-(Thiophen-2-yl)pent-3-en-2-one.**<sup>20</sup> Eluent: petroleum ether/EtOAc (100 : 1 v/v). Yellow oil, yield 28.8 mg (58%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.28 (s, 3H), 2.57 (s, 3H), 6.64 (s, 1H), 7.03–7.09 (m, 1H), 7.34 (d,  $J$  = 5.1 Hz, 1H), 7.36 (d,  $J$  = 3.4 Hz, 1H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  17.74, 32.44, 121.50, 127.43, 127.59, 128.24, 145.92, 146.23, 198.53.

**(*E*)-3-(4-(Dimethylamino)phenyl)-1-phenylbut-2-en-1-one.** Eluent: petroleum ether/EtOAc (30 : 1 v/v). Yellow solid, yield 68.6 mg (86%); m.p. 62–64 °C.  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.64 (s, 3H), 3.03 (s, 6H), 6.73 (d,  $J$  = 8.9 Hz, 2H), 7.20 (s, 1H), 7.46 (t,  $J$  = 7.4 Hz, 2H), 7.52 (d,  $J$  = 7.2 Hz, 1H), 7.56 (d,  $J$  = 8.8 Hz, 2H), 7.99 (d,  $J$  = 7.2 Hz, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  18.39, 40.39, 111.89, 118.05, 127.91, 128.24, 128.54, 129.55, 132.14, 140.44, 151.42, 156.00, 191.73. HR-MS:  $m/z$  266.15475  $[M + H]^+$ ; calcd for  $C_{18}H_{20}NO$ , 266.15394.

**(*E*)-4-(4-Methoxyphenyl)-1-phenylbut-2-en-1-one.**<sup>22</sup> Eluent: petroleum ether/EtOAc (40 : 1 v/v). Light yellow oil, yield 69 mg (90%).  $^1H$  NMR (400 MHz,  $CDCl_3$ ):  $\delta$  2.60 (d,  $J$  = 1.0 Hz, 3H), 3.86 (s, 3H), 6.95 (d,  $J$  = 8.8 Hz, 2H), 7.15–7.19 (m, 1H), 7.47 (t,  $J$  = 7.4 Hz, 2H), 7.51–7.60 (m, 3H), 7.96–8.03 (m, 2H).  $^{13}C$  NMR (101 MHz,  $CDCl_3$ ):  $\delta$  18.78, 55.49, 114.05, 120.42, 128.03, 128.31, 128.61, 132.47, 134.93, 139.78, 154.97, 160.71, 191.85.

**(*E*)-1-Phenyl-3-(*p*-tolyl)but-2-en-1-one.**<sup>22</sup> Eluent: petroleum ether/EtOAc (40 : 1 v/v). Yellow oil, yield 63 mg (89%).  $^1H$  NMR

(400 MHz, CDCl<sub>3</sub>):  $\delta$  2.40 (s, 3H), 2.60 (d,  $J$  = 1.1 Hz, 3H), 7.15–7.19 (m, 1H), 7.23 (d,  $J$  = 8.1 Hz, 2H), 7.43–7.51 (m, 4H), 7.55 (t,  $J$  = 7.2 Hz, 1H), 7.96–8.03 (m, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  18.92, 21.38, 121.42, 126.56, 128.38, 128.65, 129.44, 132.58, 139.51, 139.64, 139.93, 155.33, 192.01.

**(E)-1-Phenyl-3-(4-(trifluoromethyl)phenyl)but-2-en-1-one.** Eluent: petroleum ether/EtOAc (40:1 v/v). Yellow oil, yield 62 mg (71%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.58 (d,  $J$  = 1.1 Hz, 3H), 7.14–7.19 (m, 1H), 7.49 (t,  $J$  = 7.5 Hz, 2H), 7.58 (t,  $J$  = 7.3 Hz, 1H), 7.64–7.71 (m, 4H), 7.98–8.02 (m, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  18.97, 123.87, 124.11 (q,  $J$  = 273.2 Hz), 125.73 (q,  $J$  = 3.7 Hz), 126.97, 128.47, 128.80, 131.00 (q,  $J$  = 32.7 Hz), 133.02, 139.01, 146.43, 153.10, 191.86. HR-MS:  $m/z$  291.09991 [M + H]<sup>+</sup>; calcd for C<sub>17</sub>H<sub>14</sub>F<sub>3</sub>O, 291.09913.

**3-(4-(Dimethylamino)phenyl)cyclopent-2-en-1-one.**<sup>9a</sup> Eluent: petroleum ether/EtOAc (5:1 v/v). White solid, yield 53.2 mg (88%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.51–2.58 (m, 2H), 2.97–3.03 (m, 2H), 3.05 (s, 6H), 6.40 (s, 1H), 6.70 (d,  $J$  = 9.0 Hz, 2H), 7.56 (d,  $J$  = 9.0 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  28.50, 35.25, 40.18, 111.52, 121.51, 122.98, 128.67, 152.38, 174.67, 209.51.

**3-(4-Methoxyphenyl)cyclopent-2-en-1-one.**<sup>9a,23</sup> Eluent: petroleum ether/EtOAc (10:1 v/v). Light yellow solid, yield 46.2 mg (82%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.53–2.62 (m, 2H), 2.98–3.06 (m, 2H), 3.87 (s, 3H), 6.48 (t,  $J$  = 1.5 Hz, 1H), 6.97 (d,  $J$  = 8.8 Hz, 2H), 7.63 (d,  $J$  = 8.8 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  28.71, 35.34, 55.56, 114.37, 125.59, 126.82, 128.72, 162.20, 173.77, 209.47.

**3-(p-Tolyl)cyclopent-2-en-1-one.**<sup>9a</sup> Eluent: petroleum ether/EtOAc (10:1 v/v). White solid, yield 39 mg (75%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.41 (s, 3H), 2.53–2.61 (m, 2H), 2.98–3.08 (m, 2H), 6.54 (t,  $J$  = 1.6 Hz, 1H), 7.26 (d,  $J$  = 8.0 Hz, 2H), 7.55 (t,  $J$  = 8.2 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  21.65, 28.70, 35.34, 126.72, 126.91, 129.72, 131.41, 141.95, 174.16, 209.57.

**3-(4-(Trifluoromethyl)phenyl)cyclopent-2-en-1-one.**<sup>9b</sup> Eluent: petroleum ether/EtOAc (10:1 v/v). White solid, yield 33.8 mg (50%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.58–2.68 (m, 2H), 3.03–3.13 (m, 2H), 6.65 (d,  $J$  = 1.6 Hz, 1H), 7.72 (d,  $J$  = 8.4 Hz, 2H), 7.77 (d,  $J$  = 8.4 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  28.82, 35.44, 123.81 (q,  $J$  = 273.4 Hz), 126.03 (q,  $J$  = 3.7 Hz), 127.18, 129.52, 132.73 (q,  $J$  = 32.8 Hz), 137.50, 171.94, 208.98.

**3-(Furan-2-yl)cyclopent-2-en-1-one.**<sup>23</sup> Eluent: petroleum ether/EtOAc (10:1 v/v). Yellow solid, yield 20 mg (45%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.46–2.61 (m, 2H), 2.88–3.02 (m, 2H), 6.40 (s, 1H), 6.55 (d,  $J$  = 1.6 Hz, 1H), 6.84 (d,  $J$  = 2.5 Hz, 1H), 7.60 (s, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  27.21, 34.68, 112.54, 113.73, 124.81, 145.79, 150.59, 161.50, 208.85.

**3-(Thiophen-2-yl)cyclopent-2-en-1-one.**<sup>23</sup> Eluent: petroleum ether/EtOAc (10:1 v/v). White solid, yield 43.8 mg (89%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  2.49–2.65 (m, 2H), 2.97–3.11 (m, 2H), 6.36 (s, 1H), 7.14 (dd,  $J$  = 4.0, 4.8 Hz, 1H), 7.45 (d,  $J$  = 3.6 Hz, 1H), 7.54 (d,  $J$  = 4.8 Hz, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  29.39, 35.13, 125.88, 128.44, 128.61, 130.34, 138.73, 166.71, 208.61.

**Ethyl (E)-3-(4-(dimethylamino)phenyl)but-2-enoate.**<sup>24</sup> Eluent: petroleum ether/EtOAc (60:1 v/v). White solid, yield 65 mg

(93%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.31 (t,  $J$  = 7.1 Hz, 3H), 2.57 (s, 3H), 2.99 (s, 6H), 4.20 (q,  $J$  = 7.1 Hz, 2H), 6.12 (s, 1H), 6.68 (d,  $J$  = 8.9 Hz, 2H), 7.45 (d,  $J$  = 8.9 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.56, 17.32, 40.39, 59.64, 111.83, 112.89, 127.54, 129.09, 151.20, 155.45, 167.64.

**Ethyl (E)-(4-methoxyphenyl)but-2-enoate.**<sup>24</sup> Eluent: petroleum ether/EtOAc (200:1 v/v). Colorless oil, yield 54.9 mg (83%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.31 (t,  $J$  = 7.1 Hz, 3H), 2.56 (d,  $J$  = 1.2 Hz, 3H), 3.83 (s, 3H), 4.20 (q,  $J$  = 7.1 Hz, 2H), 6.08–6.13 (m, 1H), 6.89 (d,  $J$  = 8.9 Hz, 2H), 7.46 (d,  $J$  = 8.9 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.50, 17.78, 55.46, 59.85, 113.93, 115.43, 127.79, 134.44, 155.01, 160.53, 167.22.

**Ethyl (E)-3-(p-tolyl)but-2-enoate.**<sup>21</sup> Eluent: petroleum ether/EtOAc (200:1 v/v). Colorless oil, yield 52 mg (85%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.31 (t,  $J$  = 7.1 Hz, 3H), 2.36 (s, 3H), 2.57 (d,  $J$  = 1.2 Hz, 2H), 4.21 (q,  $J$  = 7.1 Hz, 2H), 6.10–6.15 (m, 1H), 7.18 (d,  $J$  = 8.0 Hz, 2H), 7.39 (d,  $J$  = 8.2 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.48, 17.92, 21.30, 59.88, 116.39, 126.33, 129.30, 139.24, 155.53, 167.12.

**Ethyl 3-(4-(trifluoromethyl)phenyl)but-2-enoate.**<sup>21,24</sup> Eluent: petroleum ether/EtOAc (200:1 v/v). Colorless oil, (E)/(Z) ratio 45:1, yield 57.7 mg (74%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.33 (t,  $J$  = 7.1 Hz, 3H), 2.58 (d,  $J$  = 1.3 Hz, 3H), 4.23 (q,  $J$  = 7.1 Hz, 2H), 6.11–6.18 (m, 1H), 7.57 (d,  $J$  = 8.4 Hz, 2H), 7.63 (d,  $J$  = 8.3 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.42, 18.07, 60.25, 119.08, 124.09 (q,  $J$  = 273.4 Hz), 125.60 (q,  $J$  = 3.7 Hz), 126.80, 130.91 (q,  $J$  = 32.8 Hz), 145.92, 153.91, 166.55.

**Ethyl (E)-3-(o-tolyl)but-2-enoate.** Eluent: petroleum ether/EtOAc (200:1 v/v). Colorless oil, yield 56.5 mg (92%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.31 (t,  $J$  = 7.1 Hz, 3H), 2.29 (s, 3H), 2.45 (d,  $J$  = 1.1 Hz, 3H), 4.21 (q,  $J$  = 7.1 Hz, 2H), 5.72–5.79 (m, 1H), 7.07 (d,  $J$  = 7.1 Hz, 1H), 7.12–7.24 (m, 3H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.44, 19.83, 20.92, 59.94, 119.51, 125.83, 127.19, 127.79, 130.50, 133.98, 144.02, 158.38, 166.79.

**Ethyl (E)-3-(furan-2-yl)but-2-enoate.**<sup>21</sup> Eluent: petroleum ether/EtOAc (200:1 v/v). Yellow oil, yield 37.6 mg (69%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.31 (t,  $J$  = 7.1 Hz, 3H), 2.46 (d,  $J$  = 1.1 Hz, 3H), 4.20 (q,  $J$  = 7.1 Hz, 2H), 6.35–6.39 (m, 1H), 6.46 (dd,  $J$  = 3.4, 1.8 Hz, 1H), 6.64 (d,  $J$  = 3.4 Hz, 1H), 7.44 (d,  $J$  = 1.4 Hz, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.49, 14.86, 59.91, 111.36, 112.10, 112.61, 142.24, 144.00, 154.52, 167.34.

**Ethyl (E)-3-(thiophen-2-yl)but-2-enoate.**<sup>21</sup> Eluent: petroleum ether/EtOAc (200:1 v/v). Yellow oil, yield 48 mg (82%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  1.32 (t,  $J$  = 7.1 Hz, 2H), 2.61 (d,  $J$  = 1.0 Hz, 2H), 4.20 (q,  $J$  = 7.1 Hz, 1H), 6.22–6.29 (m, 1H), 7.05 (t,  $J$  = 4.4 Hz, 1H), 7.32 (d,  $J$  = 4.4 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  14.49, 17.43, 60.01, 114.41, 126.83, 127.19, 128.06, 145.72, 147.92, 166.90.

**Ethyl 4'-methoxy-3,4,5,6-tetrahydro-[1,1'-biphenyl]-2-carboxylate.**<sup>25</sup> Eluent: petroleum ether/EtOAc (100:1 v/v). Colorless oil, yield 60 mg (77%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>):  $\delta$  0.91 (t,  $J$  = 7.1 Hz, 3H), 1.66–1.80 (m, 4H), 2.32–2.47 (m, 4H), 3.80 (s, 3H), 3.91 (q,  $J$  = 7.1 Hz, 2H), 6.84 (d,  $J$  = 8.7 Hz, 2H), 7.08 (d,  $J$  = 8.7 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>):  $\delta$  13.85, 22.08, 22.68, 26.90, 32.62, 55.36, 60.16, 113.47, 127.92, 128.16, 135.87, 144.82, 158.82, 170.51.

**Ethyl 4'-methyl-3,4,5,6-tetrahydro-[1,1'-biphenyl]-2-carboxylate.** Eluent: petroleum ether/EtOAc (100 : 1 v/v). Light yellow oil, yield 54.5 mg (74%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 0.88 (t, *J* = 7.1 Hz, 3H), 1.66–1.80 (m, 4H), 2.33 (s, 1H), 2.34–2.46 (m, 4H), 3.89 (q, *J* = 7.1 Hz, 2H), 7.03 (d, *J* = 7.9 Hz, 2H), 7.11 (d, *J* = 8.0 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 13.74, 21.28, 22.09, 22.66, 26.80, 32.70, 60.14, 126.86, 127.92, 128.76, 136.71, 140.63, 145.55, 170.30. HR-MS: *m/z* 245.15282 [M + H]<sup>+</sup>; calcd for C<sub>16</sub>H<sub>21</sub>O<sub>2</sub>, 245.15361.

**Ethyl 4'-(dimethylamino)-3,4,5,6-tetrahydro-[1,1'-biphenyl]-2-carboxylate.** Eluent: petroleum ether/EtOAc (60 : 1 v/v). Light yellow oil, yield 44 mg (54%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 0.94 (t, *J* = 7.1 Hz, 3H), 1.64–1.81 (m, 4H), 2.32–2.46 (m, 4H), 2.93 (s, 6H), 3.94 (q, *J* = 7.1 Hz, 2H), 6.66 (d, *J* = 8.7 Hz, 2H), 7.05 (d, *J* = 8.8 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 13.97, 22.17, 22.80, 27.10, 32.27, 40.75, 60.13, 112.15, 127.03, 127.91, 131.38, 144.96, 149.96, 171.10. HR-MS: *m/z* 274.17941 [M + H]<sup>+</sup>; calcd for C<sub>17</sub>H<sub>24</sub>N<sub>2</sub>O<sub>2</sub>, 274.18016.

**4-(4-(Dimethylamino)phenyl)-2H-chromen-2-one.**<sup>9d</sup> Eluent: petroleum ether/EtOAc (30 : 1 v/v). Yellow solid, yield 70.2 mg (88%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 3.06 (s, 6H), 6.34 (s, 1H), 6.81 (d, *J* = 8.8 Hz, 2H), 7.20–7.26 (m, 1H), 7.39 (d, *J* = 8.9 Hz, 3H), 7.49–7.57 (m, 1H), 7.70 (dd, *J* = 8.0, 1.2 Hz, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 31.10, 40.39, 112.02, 113.48, 117.46, 119.47, 122.51, 124.01, 127.38, 129.97, 131.65, 151.48, 154.48, 155.91, 161.49.

**4-(4-Methoxyphenyl)-2H-chromen-2-one.**<sup>9e</sup> Eluent: petroleum ether/EtOAc (60 : 1 v/v). White solid, yield 59 mg (78%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 3.90 (s, 3H), 6.36 (s, 1H), 7.05 (d, *J* = 8.8 Hz, 2H), 7.20–7.26 (m, 1H), 7.37–7.45 (m, 3H), 7.51–7.59 (m, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 55.57, 114.42, 114.71, 117.45, 119.22, 124.21, 127.13, 127.52, 130.07, 131.93, 154.32, 155.45, 160.92, 161.06.

**4-(*p*-Tolyl)-2H-chromen-2-one.**<sup>9e</sup> Eluent: petroleum ether/EtOAc (60 : 1 v/v). White solid, yield 50 mg (70%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 2.46 (s, 3H), 6.37 (s, 1H), 7.20–7.25 (m, 1H), 7.30–7.39 (m, 4H), 7.41 (d, *J* = 8.0 Hz, 1H), 7.51–7.59 (m, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 21.51, 115.02, 117.44, 119.21, 124.23, 127.18, 128.53, 129.68, 131.96, 132.43, 140.03, 154.32, 155.87, 161.02.

**4-(4-(Trifluoromethyl)phenyl)-2H-chromen-2-one.**<sup>9e</sup> Eluent: petroleum ether/EtOAc (60 : 1 v/v). White solid, yield 63.3 mg (73%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 6.40 (s, 1H), 7.22–7.31 (m, 1H), 7.39 (dd, *J* = 8.0, 1.3 Hz, 1H), 7.44 (d, *J* = 7.8 Hz, 1H), 7.55–7.63 (m, 3H), 7.82 (d, *J* = 8.1 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 115.93, 117.66, 118.61, 123.88 (q, *J* = 273.3 Hz), 124.57, 126.09 (q, *J* = 3.7 Hz), 126.70, 129.04, 131.95 (q, *J* = 32.9 Hz), 132.48, 138.85, 154.29, 160.39.

**4-(Furan-2-yl)-2H-chromen-2-one.**<sup>9d</sup> Eluent: petroleum ether/EtOAc (60 : 1 v/v). Yellow solid, yield 35.1 mg (55%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 6.65 (s, 1H), 6.69 (s, 1H), 7.06 (d, *J* = 3.2 Hz, 1H), 7.33 (t, *J* = 7.8 Hz, 1H), 7.40 (d, *J* = 8.3 Hz, 1H), 7.58 (t, *J* = 7.7 Hz, 1H), 7.71 (s, 1H), 8.17 (d, *J* = 8.0 Hz, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 111.65, 112.51, 114.59, 116.60, 117.65, 124.44, 126.46, 131.98, 142.05, 145.41, 148.52, 154.37, 161.02.

**4-(Thiophen-2-yl)-2H-chromen-2-one.**<sup>9d</sup> Eluent: petroleum ether/EtOAc (60 : 1 v/v). Yellow solid, yield 47.5 mg (69%). <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 6.51 (s, 1H), 7.24 (dd, *J* = 5.0, 3.7 Hz, 1H), 7.28–7.35 (m, 1H), 7.38–7.47 (m, 2H), 7.54–7.64 (m, 2H), 7.94 (dd, *J* = 8.0, 1.1 Hz, 1H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 115.21, 117.62, 118.40, 124.46, 126.75, 128.27, 128.66, 129.49, 132.26, 136.02, 148.12, 154.29, 160.57.

**3-(4-(Dimethylamino)phenyl)-*N,N*-dimethylbut-2-enamide.** Eluent: petroleum ether/EtOAc (5 : 1 v/v). Light yellow oil, (*E*)/(*Z*) ratio 10 : 1, yield 70 mg (99%). The NMR spectroscopic data for the major isomer are given. <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 2.14 (s, 3H), 2.64 (s, 3H), 2.84 (s, 3H), 2.96 (s, 6H), 5.78 (s, 1H), 6.64 (d, *J* = 8.8 Hz, 2H), 7.20 (d, *J* = 8.8 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 24.10, 34.40, 37.81, 40.39, 111.77, 118.47, 127.83, 128.02, 142.28, 150.19, 170.04. HR-MS: *m/z* 233.16545 [M + H]<sup>+</sup>; calcd for C<sub>14</sub>H<sub>20</sub>N<sub>2</sub>O, 233.16484.

**3-(4-Methoxyphenyl)-*N,N*-dimethylbut-2-enamide.**<sup>26</sup> Eluent: petroleum ether/EtOAc (5 : 1 v/v). Colorless oil, (*E*)/(*Z*) ratio 4 : 1, yield 66 mg (99%). The NMR spectroscopic data for the major isomer are given. <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 2.15 (s, 3H), 2.65 (s, 3H), 2.82 (s, 3H), 3.81 (s, 3H), 5.88 (s, 1H), 6.84 (d, *J* = 8.7 Hz, 2H), 7.25 (d, *J* = 8.7 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 24.29, 34.35, 37.72, 55.24, 113.58, 119.96, 128.28, 132.52, 142.05, 159.36, 169.43.

***N,N*-Dimethyl-3-(*p*-tolyl)but-2-enamide.** Eluent: petroleum ether/EtOAc (5 : 1 v/v). Light yellow oil, (*E*)/(*Z*) ratio 2 : 1, yield 54.6 mg (90%). The NMR spectroscopic data for the major isomer are given. <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>): δ 2.16 (d, *J* = 1.3 Hz, 3H), 2.34 (s, 3H), 2.64 (s, 3H), 2.81 (s, 3H), 5.91 (s, 1H), 7.11 (d, *J* = 8.0 Hz, 2H), 7.20 (d, *J* = 8.4 Hz, 2H). <sup>13</sup>C NMR (101 MHz, CDCl<sub>3</sub>): δ 21.24, 24.32, 34.36, 37.72, 120.37, 126.84, 128.95, 137.25, 137.85, 142.63, 169.30. HR-MS: *m/z* 204.13870 [M + H]<sup>+</sup>; calcd for C<sub>13</sub>H<sub>18</sub>NO, 204.13829.

## Conflicts of interest

There are no conflicts to declare.

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